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Math 400

Section 2.1 The Tangent and Velocity Problems

Objective 1: Average velocity.

$$Distance = Rate \cdot Time$$

$$Rate = \frac{Distance}{Time}$$

So if you traveled 200 miles in 4 hours, your rate (average velocity) would be

$$Average\ Velocity(Rate) = \frac{Distance}{Time} = \frac{200\ miles}{4\ hours} = 50\ miles\ per\ hour$$

Example 1: A rock is launched vertically upward from the ground with a speed of 96 ft/s. Neglecting air resistance, a well-known formula from physics states that the position of the rock after *t* seconds is given by the function

$$s(t) = -16t^2 + 96t.$$

The position *s* is measured in feet with $s = 0$ corresponding to the ground. **Find the average velocity of the rock between each pair of times.** (Section 2.1 page 54 in Calculus Early Transcendentals 2nd edition by Briggs and Cochran)

(a.) $t = 1\ second$ and $t = 3\ seconds$

$$v_{av} = \frac{change\ in\ position}{time\ elapsed}$$

$$= \frac{s(3) - s(1)}{3 - 1}$$

$$= \frac{\left(-16(3)^2 + 96(3)\right) - \left(-16(1)^2 + 96(1)\right)}{2}$$

$$= \frac{144 - 80}{2}$$

$$= \frac{64\ ft}{2\ s} = 32\ ft/s$$

Example 1: (continued)

(b.) $t = 1\ second$ and $t = 2\ seconds$

$$v_{av} = \frac{change\ in\ position}{time\ elapsed}$$

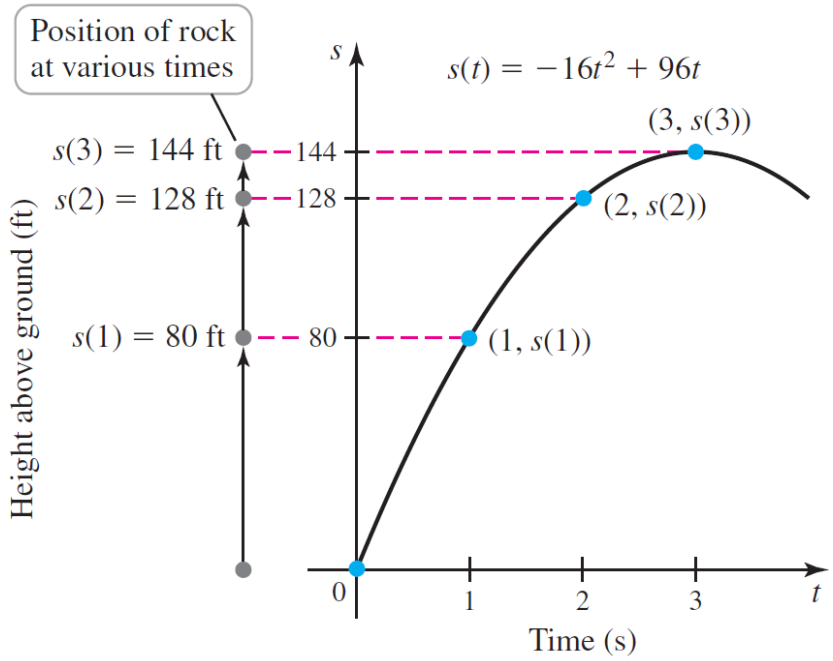
$$= \frac{s(2) - s(1)}{2 - 1}$$

$$= \frac{\left(-16(2)^2 + 96(2)\right) - \left(-16(1)^2 + 96(1)\right)}{1}$$

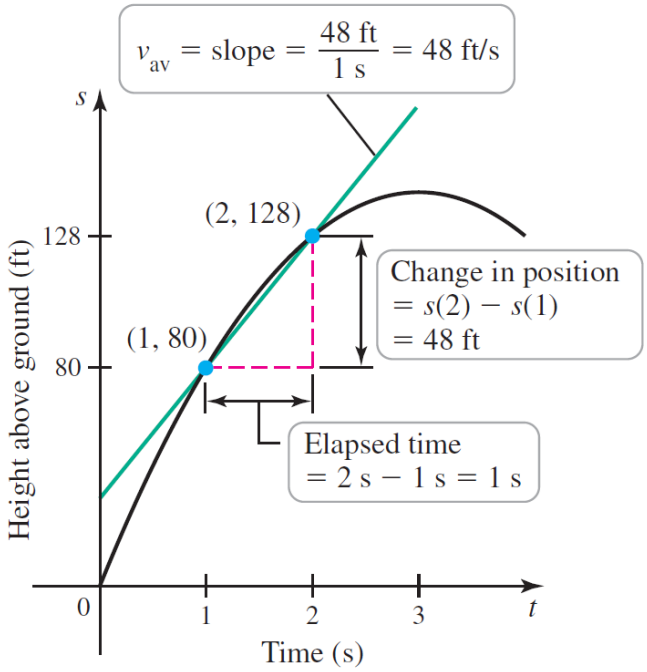
$$= \frac{128 - 80}{2}$$

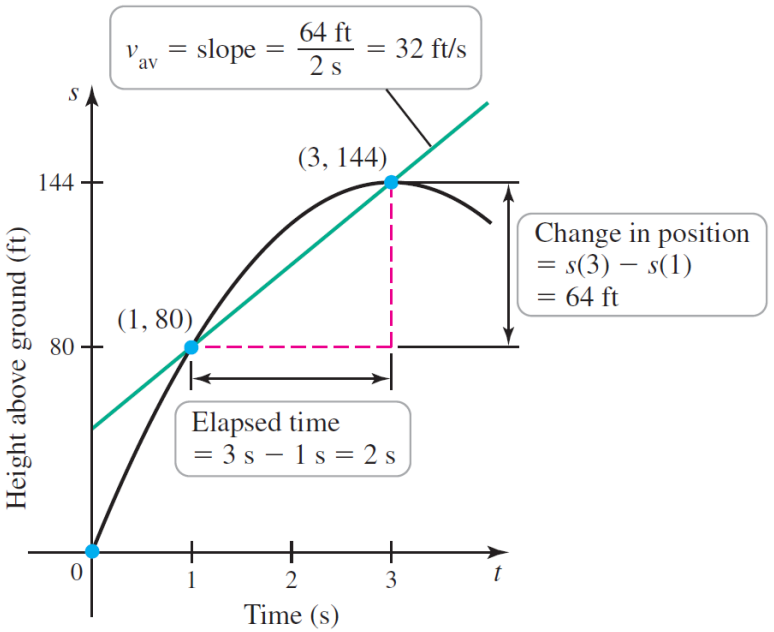
$$= \frac{48\ ft}{1\ s} = 48\ ft/s$$

Example 1: (continued)

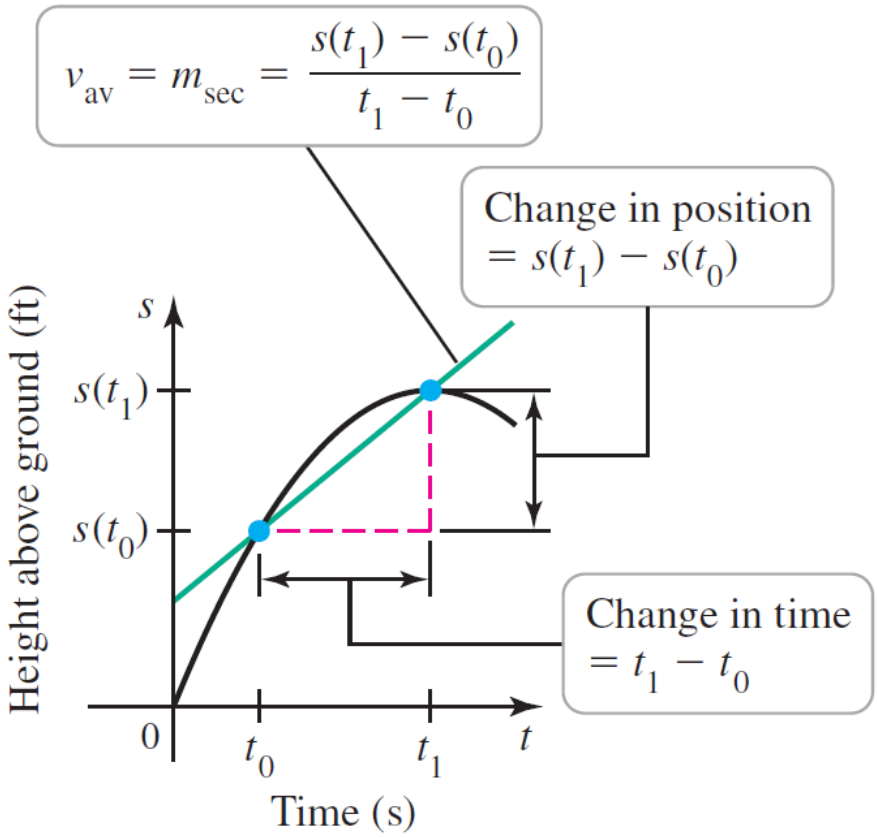


Note: The graph is not the path of the rock. The rock travels up and down on a vertical line. (Section 2.1 page 55 in Calculus Early Transcendentals 2nd edition by Briggs and Cochran)





Example 1: (continued)



Objective 2: Instantaneous velocity.

Example 2: Estimate the instantaneous velocity of the rock in Example 1 at the single point *t = 1*.

$$s(t) = -16t^2 + 96t$$

The average velocity of the rock over the time interval $[1, t]$ is given by the formula

$v_{av} = \frac{s(t)-s(1)}{t-1}$.

$$v_{av} = \frac{(-16t^2 + 96t) - \left(-16(1)^2 + 96(1)\right)}{t - 1}$$

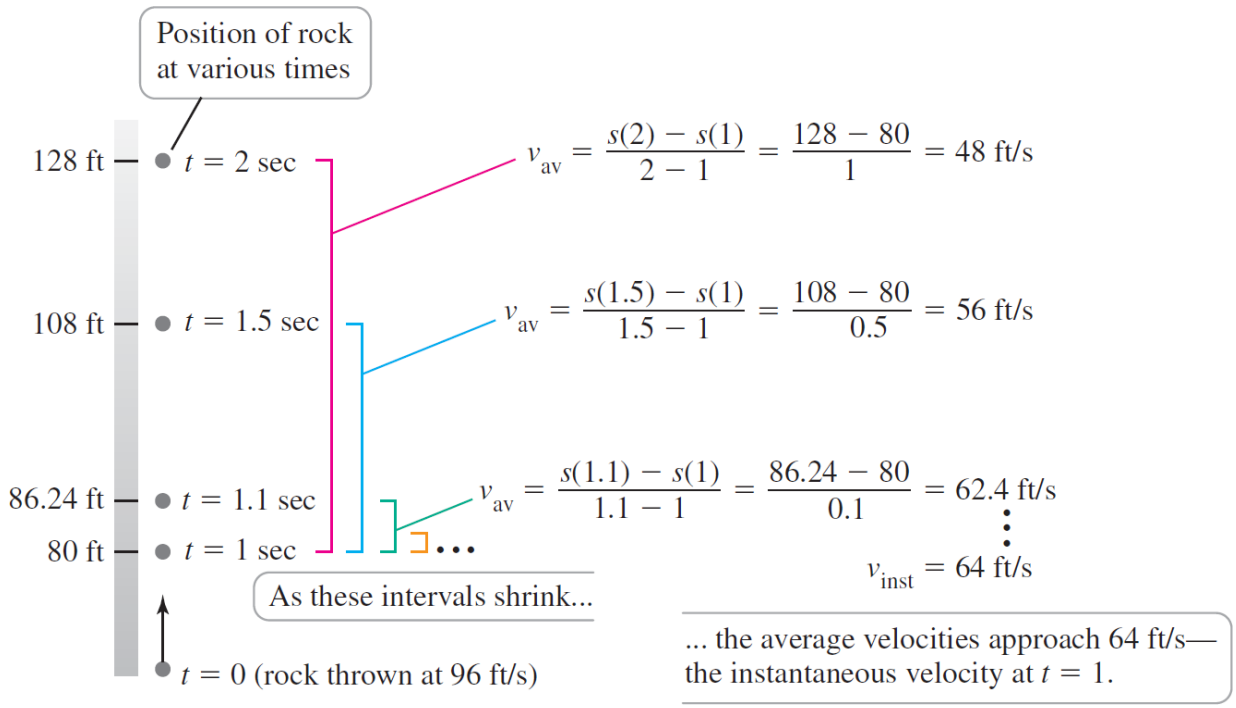
t	Time interval,$[1, t]$	$v_{av} = \frac{s(t) - s(1)}{t - 1}$
2	[1,2]	$v_{av} = \frac{\left(-16(2)^2 + 96(2)\right) - (80)}{2 - 1}$ $= 48\ ft/sec$
1.5	[1,1.5]	$v_{av} = \frac{\left(-16(1.5)^2 + 96(1.5)\right) - (80)}{1.5 - 1}$ $= 56\ ft/sec$

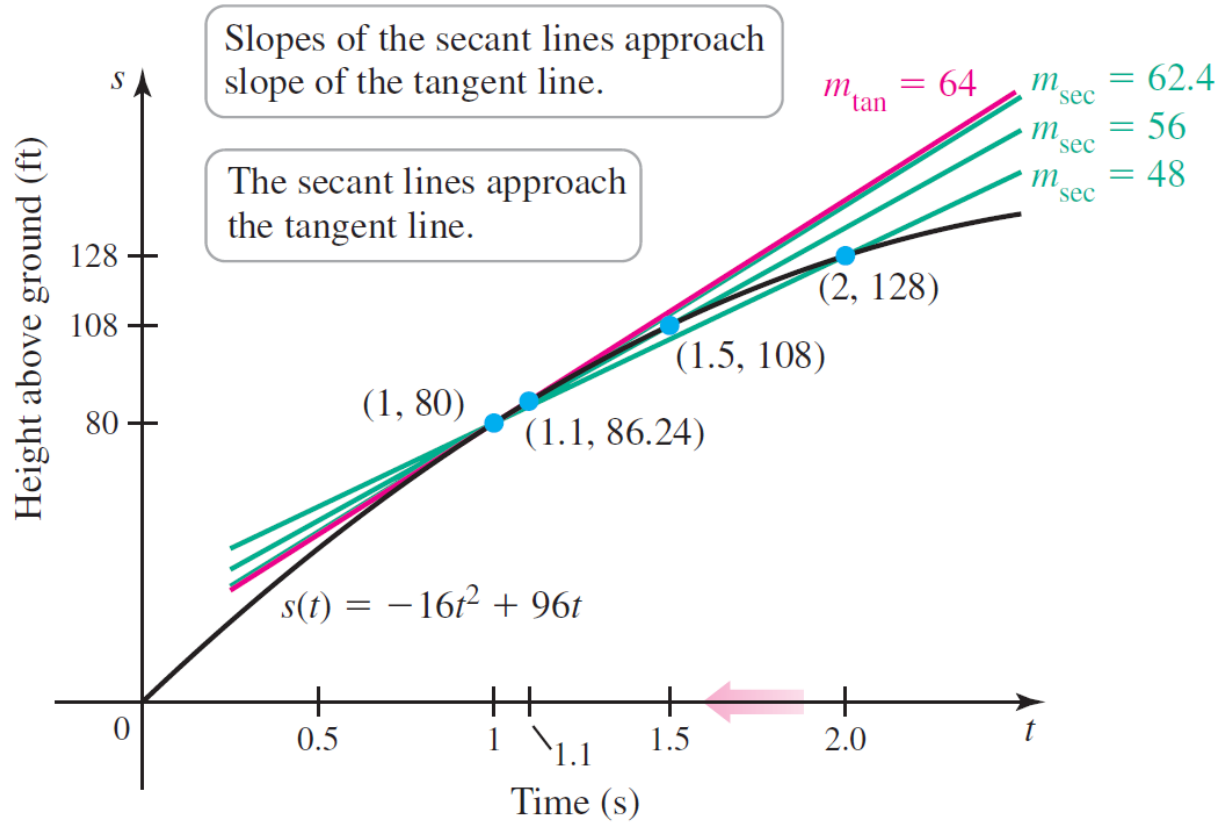
Example 2: (continued)

t	Time interval, $[1, t]$	$v_{av} = \frac{s(t) - s(1)}{t - 1}$
1.1	$[1,1.1]$	$v_{av} = \frac{\left(-16(1.1)^2 + 96(1.1)\right) - (80)}{1.1 - 1}$ $= 62.4\ ft/sec$
1.01	$[1,1.01]$	$v_{av} = \frac{\left(-16(1.01)^2 + 96(1.01)\right) - (80)}{1.01 - 1}$ $= 63.84\ \ ft/sec$
1.001	$[1,1.001]$	$v_{av} = \frac{\left(-16(1.001)^2 + 96(1.001)\right) - (80)}{1.001 - 1}$ $= 63.984\ ft/sec$

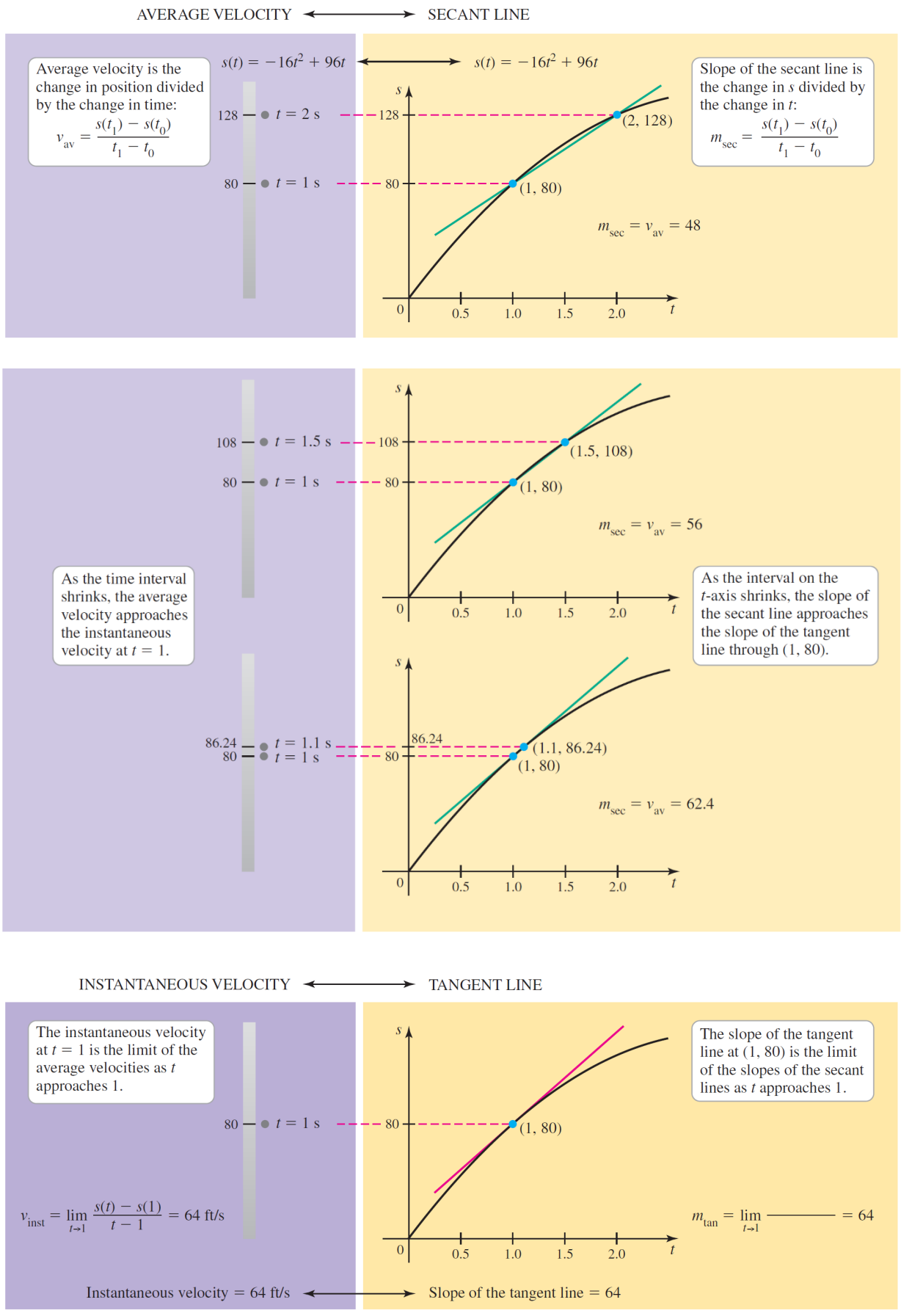
$$intantaneous\ velocity\ at\ t = 1\ second = \lim_{t \to 1^+} \frac{s(t) - s(1)}{t - 1} = 64\ ft/sec$$

Example 2: (continued)





Example 2: (continued)



Objective 3: Slope of the tangent line.

Example 3: The point $P(0.5,0)$ lies on the curve $f(x)=\cos \pi x$. (Section 2.1 page 82 #4 in Stewart's Calculus.)

(a.) If Q is the point $(x,\cos \pi x)$, find the slope of the secant line PQ (correct to six decimal places) for the following values of x.

x	$f(x)=\cos \pi x$	$(x,\cos \pi x)$	m_{PQ}
0.4			
0.49			
0.499			
0.6			
0.51			
0.501			

(b.) Use the results of part (a.) to guess the value of the slope of the tangent line to the curve at $P(0.5,0)$.

(c.) Using the slope from part (b.), find an equation of the tangent line to the curve at $P(0.5,0)$.

Example 3: The point $P(0.5,0)$ lies on the curve $f(x)=\cos \pi x$. (Section 2.1 page 82 #4 in Stewart's Calculus.)

(a.) If Q is the point $(x,\cos \pi x)$, find the slope of the secant line PQ (correct to six decimal places) for the following values of x.

x	$f(x)=\cos \pi x$	$(x,\cos \pi x)$	m_{PQ}
0.4	0.309017	(0.4,0.309017)	$\frac{f(0.4)-f(0.5)}{0.4-0.5}=-3.09170$
0.49	0.031411	(0.49,0.031411)	$\frac{f(0.49)-f(0.5)}{0.49-0.5}=-3.14076$
0.499	0.0039142	(0.499,0.0039142)	$\frac{f(0.499)-f(0.5)}{0.499-0.5}=-3.141587$
0.6	-0.309017	$(0.6,-0.309017)$	$\frac{f(0.6)-f(0.5)}{0.6-0.5}=-3.090170$
0.51	-0.031411	$(0.51,-0.031411)$	$\frac{f(0.51)-f(0.5)}{0.51-0.5}=-3.141076$
0.501	-0.003142	$(0.501,-0.003142)$	$\frac{f(0.501)-f(0.5)}{0.501-0.5}=-3.141587$

(b.) Use the results of part (a.) to guess the value of the slope of the tangent line to the curve at $P(0.5,0)$.

Example 3: (continued)

(c.) Using the slope from part (b.), find an equation of the tangent line to the curve at $P(0.5,0)$.

